

Predicting Short-Term NIFTY IT Movements Using AI Search Volumes

Abstract

This paper is an empirical study examining the relationship between public interest in Artificial Intelligence (quantified via search volumes for Gemini, DeepSeek, and ChatGPT) and the short-term price movement of the NIFTY IT Index. The model developed in this program is a generalised framework capable of identifying correlations between stock market price movements and search volume indices across arbitrary sets of keywords regardless of their number and type.

Using a Random Forest Classification framework optimized via Time Series Cross-Validation the model achieves an accuracy of 57.27% outperforming random, null and persistence baselines. However statistical significance tests such as Wilcoxon and AUROC suggest marginal significance. Overall it can be concluded that there is a weak but detectable correlation between AI Keywords and IT sector returns.

Data Acquisition

Data for this project was acquired via Google Trends with the following parameters:

Region: Worldwide

- Categories: All Categories
- Search: Web Search
- Time: Jan 27 2023 to Nov 27 2025
- Search pattern: Daily

Google Trends limits search to around 180 days data was acquired for each 180 day window and then the subsequent window was multiplied with a proportionality constant. The proportionality constant(m) was calculated in the following way:

Taking x as the last intensity of the previous window

Taking y as the first intensity of the current window

(The last day of the previous window and the first day of the current window are the same day)

If x and y are equal: $m = 1$

Else: $m = x/y$

If either x or y are equal to 0 we go back previous entries and land on a common date for both windows where neither is equal to 0.

After acquiring data for each keyword the data was normalised across all keywords by finding a common date with non zero intensity and scaling the keywords proportionally.

Feature Engineering:

To capture velocity and acceleration of trends three feature categories were chosen:

Raw Search Volume $X_{i,t}$

- Lagged Day Volume (Previous Day's intensity) $X_{i,t-1}$
- Daily Change $X_{i,t} - X_{i,t-1}$

The model was then used to predict whether the price would increase or decrease from the previous day.

Model Architecture

The model architecture utilises the Random Forest Classifier, an ensemble learning estimator that fits a correlation that fits a collection of N uncorrelated decision trees on various sub-samples of the dataset. The architecture relies on Bootstrap Aggregating (Bagging) to reduce variance without increasing bias. Each tree is constructed by using feature randomness by selecting a random subset of features at each node split. This ensures each individual tree is uncorrelated to another allowing the forest to capture complex non-linear interactions between disparate search signals—such as a simultaneous spike in "ChatGPT" interest and a dip in "Gemini" volume—that a linear model would likely miss. The final prediction is determined through a majority voting mechanism which provides a robust defense against the "noise" and outliers inherent in high-frequency Google Trends data.

To prevent look ahead bias the model utilises Time Series Split with $K = 5$ sliding windows. Unlike a standard K -fold cross validation this approach ensures that the training set precedes the test set in a chronological order hence simulating real world trading environments.

The architecture is refined through a GridSearchCV layer which performs a search over a specified hyperparameter space. This process optimizes the `max_depth` to prevent the trees from growing deep enough to "memorize" specific historical search events, and adjusts `min_samples_split` to ensure that each leaf node represents a statistically significant cluster of days.

```
--- Grid Search Results ---
Best Parameters Found:
{'max_depth': 15, 'max_features': 'sqrt', 'min_samples_split': 2,
 'n_estimators': 100}
```

Experimental Results

To compare performance we create two new models to analyze our performance. The first model is the null model which consists of the assumption of a constant and consistent increase in price. On this model we perform the same time series cross-validation test and use a Random Forest Classifier to create the model. We

analyze this model to account for class imbalance. For example, if the dataset has 60% “up” day a model with less than 60% accuracy is doing no better than making a weighted coin with probability of 0.6 and 0.4. By running the grid search on this model we can predict and the extra accuracy we get is due to the feature engineering that we chose and has an impact on the accuracy.

The second model created was the persistence model. The persistence model is chosen so that you can establish that your model is making predictions irrespective of the momentum. Ex: In a bull run you could get high accuracy by predicting “up” whenever there is an “up” momentum. If the model beats the accuracy of the persistence model it means the keywords are leading indicators; they are catching the turn in momentum before the price action does.

Metric	Full Model (Keywords)	Null Model (Constant)	Persistence (Lag-1)
Accuracy	0.5727	0.46362	0.4460
F1-Score	0.4051	0.6335	0.4350

In the above table we can see that our model has a high accuracy but a low f1-score. This means that the model has significant false positives and false negatives.

This proves that ChatGPT and Gemini search volumes are not just "reflecting" what the market already did; they are adding specific predictive value that the market price alone does not have.

However, to know the extent of the correlation between the keywords and price change we need to perform a statistical analysis. Hence below are the results of the various statistical tests performed.

1) Wilcox Signed-Rank Test

While a simple comparison of accuracy (e.g. 57% vs 46%) shows that the model performed better it doesn't prove that the improvement was intentional rather than just a result of a "lucky" data split.

The Wilcoxon test is designed specifically to test for paired datasets. Here the “Full Model” and the “Null Model” are evaluated on the exact 5 folds the test accounts for temporal variance and market regime shifts inherent in each split. This evaluates whether the inclusion of AI search features consistently provides predictive advantage across different windows of time.

This test yielded a **p-value of 0.25** which exceeds the traditional significance threshold of 0.05. In the context of financial time series such a p-value is an indicator of weak signals.

2) Area Under ROC Test (AUROC)

The AUROC test is essential because it can tell the ability to distinguish between “up” and “down” days. While accuracy would only tell whether the final guess would be correct AUROC measures quality of underlying probabilities. Having an AUROC of **0.5296** signifies that the model has a weak but non-random predictive signal. This means that the search words are contributing some information.

3) Confusion Matrix

The model’s performance was tested over 110 trading days and the results indicate a **conservative predictive bias** model. Below is the model’s confusion matrix:

	Predicted: 0 (Down)	Predicted: 1 (Up)
Actual: 0 (Down)	47	12
Actual: 1 (Up)	35	16

- 47 True Negatives - The model correctly predicted 47 "Down" days.
- 16 True Positives - The model correctly predicted 16 "Up" days.
- 12 False Positives - The model predicted “Up” but the market went “Down”
- 35 False Negatives - The model predicted “Down” but the market went “Up”

Here we can see an *asymmetric* error with two types of error:

- Type 1 error (False Positive - 12): The market predicts an “Up” day during a market decline which represents a bad trade where capital is put at risk
- Type 2 error (False Negative - 35): The model misses an “Up” day which represents a loss of opportunity but preserves capital.

Metric Interpretation:

1) Precision

The model achieves a precision of **0.5714** $TP/(TP + FP)$. A 7.14% increase over a baseline of 50% indicates that when the model issues a “buy” signal there is a statistical observable increase in probability of positive return.

2) Recall

A recall score of **0.3137** $TP/TP + FN$ means that the model was only able to capture slightly less than 1/3rd of the upward movement of the index. This low sensitivity confirms that search volume data, while informative, does not capture all market-moving catalysts such as institutional earnings reports or macro-economic policy shifts.

3) F1-score

The weighted F1-score of **0.4051** reflects a model that is heavily weighted toward identifying market "Down" days compared to "Up" days .

	Support
Class 0	59
Class 1	51

Feature Importance

Simply inspecting which features appear in a trained model is not sufficient to understand *why* the model makes its predictions because all inputs do not contribute equally to reducing uncertainty. **Gini importance** addresses this limitation by explicitly quantifying how much each keyword reduces impurity across the decision trees in the optimized Random Forest making it a meaningful measure of feature influence rather than mere presence.

Over the analyzed period the extracted importances show that **ChatGPT- and Gemini-related metrics dominate the model's decision-making** together capturing the largest share of explanatory power with individual weights ranging from **13.7% to 16.1%**. In contrast DeepSeek-related variables contribute less than 4% indicating a comparatively weak role in predicting market direction. This imbalance suggests that mainstream AI sentiment—reflected through ChatGPT and Gemini search volume—currently exerts a far stronger influence on Nifty IT index movements.

```
--- Feature Importances (Optimized Model) ---
ChatGPT: (Worldwide)_Change      0.161316
Gemini: (Worldwide)               0.160004
ChatGPT: (Worldwide)              0.156781
ChatGPT: (Worldwide)_Lag1         0.153324
Gemini: (Worldwide)_Lag1          0.144030
Gemini: (Worldwide)_Change        0.137258
DeepSeek: (Worldwide)_Lag1        0.037122
DeepSeek: (Worldwide)             0.029075
DeepSeek: (Worldwide)_Change      0.021089
```

Checking Economic Baseline Return

A simulation of a “long only” algorithmic strategy in which we take position only when the market is expected to rise, otherwise staying and holding the cash. It avoids short

selling in the market which limits downside from unlimited losses but also sacrifices profits during sustained market declines.

To evaluate performance we compute **cumulative returns** using the cumulative product which captures the compounded growth of an investment over time and allows a direct comparison between the market and the model-driven strategy.

The markets net return is calculated by:

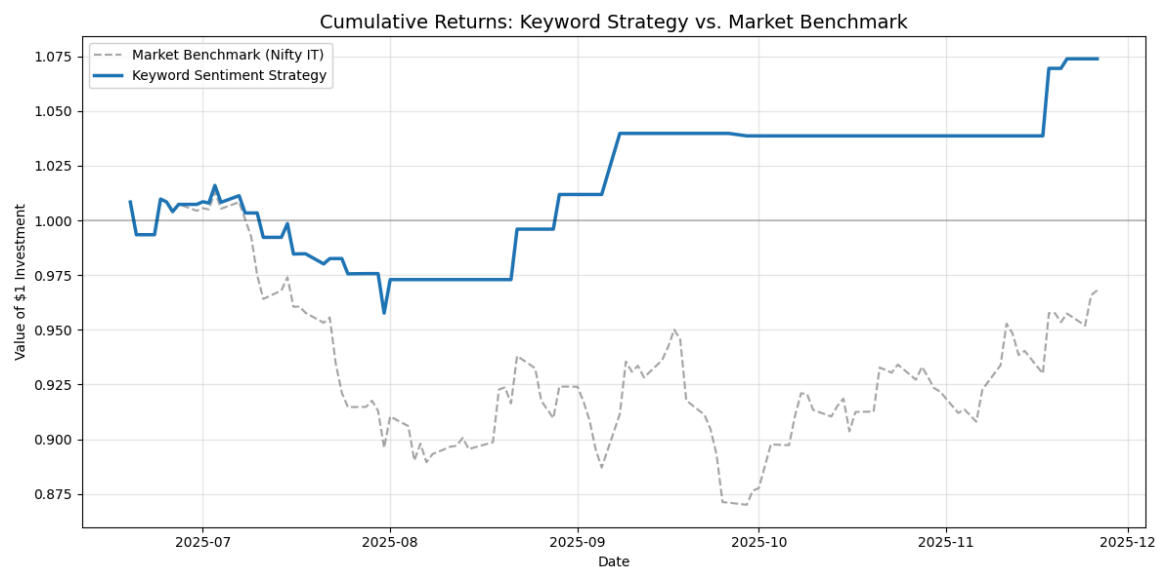
$$\text{Market_return} = \prod (1 + \text{daily_return})$$

The strategy return was calculated using:

$$\text{strategy_return} = \prod (1 + \text{daily_return} * \text{prediction})$$

Where prediction is 1 if the model believes an upward price movement and 0 otherwise. This effectively places the portfolio in the market only on days with a positive signal and keeps it in cash on all other days.

Using this long-only signal-driven approach tested on the final data fold the market produced a cumulative return of **0.968** whereas the model-based strategy achieved a higher cumulative return of **1.074** indicating that selective market participation improved overall performance during the evaluation period.



However, simply comparing cumulative returns over a fixed time period is **not sufficient** to judge the quality of a trading strategy simply because higher returns can come from taking disproportionately higher risk. The **Sharpe ratio** addresses this limitation by

measuring *risk-adjusted performance*. This makes it a good indicator comparing a model-driven strategy to a passive market benchmark.

The strategy achieved an **annualized Sharpe ratio of 1.6975** indicating strong and consistent excess returns relative to the risk over the tested time horizon. This is in contrast to the **market Sharpe ratio of -0.3275** which shows that the market delivered negative risk-adjusted performance meaning investors were not compensated for the volatility they endured. This result is reinforced by the volatility comparison: the strategy exhibited **significantly lower annualized volatility (9.91%)** than the market (**17.91%**) demonstrating that the model not only improved returns but also substantially reduced risk through selective market participation.

```
--- Risk-Adjusted Performance (Sharpe Ratio) ---  
Annualized Strategy Sharpe Ratio: 1.6975  
Annualized Market Sharpe Ratio: -0.3275  
Strategy Annualized Volatility: 9.91%  
Market Annualized Volatility: 17.91%
```

Conclusion:

The intent of this study was to find whether public interest in Artificial Intelligence - proxied through Google Search volumes for popular LLM's such as ChatGPT, Gemini and DeepSeek contain enough information to predict short term movements in the NIFTY IT index.

The Random Forest model consistently outperformed random, null, and persistence model baselines in terms of accuracy which demonstrates that the keywords provide incremental predictive value beyond simple price momentum or class imbalance effects. However statistical tests such as the Wilcoxon signed-rank test and AUROC confirm that this signal is weak which aligns with broad literature on alternative data in financial markets where most non-price signals demonstrate low signal-to-noise ratios.

Feature importance analysis further clarifies the nature of this relationship. Search activity related to mainstream AI platforms—ChatGPT and Gemini—dominates the model's explanatory power while DeepSeek-related metrics contribute minimally

Despite having only 57% accuracy the strategy was able to achieve higher cumulative returns with lower volatility resulting in a much higher Sharpe ratio. Overall the findings suggest that AI-related search trends should not be seen as leading forces of market direction but as complimentary signals that can enhance performance when integrated into signal filtering strategies.

More work can be done to extend this research by incorporating additional data sources other financial markets or by comparing with respect to different indicators in different markets (Example: virus, pandemic, illness vs healthcare indices). Further work can be done to incorporate intraday information to isolate conditions under which public attention affects market prices.